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Editor-in-Chief
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Dear Editor,

We are pleased to submit our manuscript, "The Joule Point: an Energy-Optimal Operating Point for AI Inference" by Alexander Apartsin and Yehudit Aperstein, for consideration in Sustainable Computing: Informatics and Systems.

The paper shows by direct measurement that the standard practice of running data-center GPUs at full rated power wastes a quarter to a third of the energy spent per inference. For 20 inference models on multiple NVIDIA GPU classes, board power follows a superlinear response law over the operating point (GPU, power cap), so energy per inference is U-shaped with a minimum, the Joule Point, at 43 to 46 per cent of rated power. Under load this optimum is nearly a per-card constant, so a single static cap per card type, set once, captures the saving that prior energy-aware systems search for per job at runtime. A trace-driven fleet simulation over the measured curves serves equal work for 18 to 45 per cent less energy under a power budget, and an incentive analysis locates the obstacle to adoption in per-hour GPU pricing rather than in the technology.

The work sits squarely in the journal's scope of energy-efficient computing systems and green data centers. Alongside the paper we release ELF, the measurement dataset behind every figure and number, together with the exact measurement harnesses, archived at Zenodo ([doi:10.5281/zenodo.22058568](https://doi.org/10.5281/zenodo.22058568)) under CC-BY-4.0 (data) and MIT (code).

This manuscript is original, has not been published previously, and is not under consideration by any other journal. All authors have approved the manuscript and agree with its submission. The authors declare no competing interests.

Thank you for considering our submission.

Sincerely,

Yehudit Aperstein (corresponding author)
on behalf of both authors